

Kronos Project — Failed-Strategies Postmortem

Every directional / scalping approach we tried, why it died, with the numbers

Compiled 2026-06-27 · Testnet / backtest figures throughout — no real capital was risked.

One root cause killed almost everything. At the frequency we traded, the directional signal was always thinner than the cost of trading on it. Round-trip cost on Binance is **~15 bps with the BNB discount, ~20 without** — every fast edge we found was worth a few bps per trade, *smaller than the toll*. We proved this at least five independent ways (Kronos, meta-labeler, consensus TA, the entry-rule search, OFI). **Nothing in the directional or market-timing family survived the 5-year, multi-regime bar.** Two slower ideas lasted longest: intraday-TSM (base failed 5yr; a selective 'careful bets' version failed too — pickier was *worse* out-of-sample) and stablecoin-flow (now also failed: **-8.4% vs buy-and-hold +40%**). The one family never disproved — because never built — is the structural **'be the house'** set (DeFi LP vaults, vol risk premium): collect a premium instead of predicting, and skip the cost wall entirely.

Master scorecard

Strategy	What it was	Best honest result	Verdict
Kronos (core)	Kronos-small ML price predictor, 30x Monte-Carlo, edge gate ≥ 0.53	Take-all 47–49% win, Sharpe -0.04 to -0.09 (negative before fees)	DEAD — never cracked 50%
Kelly / meta-labeler	Logistic-regression P(win) filter over journal features	v1 33.3% holdout vs 61.9% majority baseline; retrain $\sim 45\%$ vs 51%	DEAD — anti-predictive; never left shadow
Entry-rule search	Mined 2–3 condition combos from 2.7k–4.7k trades	4 'greens', OOS Sharpe 0.07–0.27 on ~ 30 –130 trades, never replicated	DEAD — data-mining noise
Consensus TA board	7-indicator trend net-bias (MACD, Supertrend, Stoch, CCI, Boll, Donchian, OBV)	48% win at 1:1 over 5 months	DEAD — lost money with the signal
RSI-2 reversion	Connors fade-the-extreme (counter-test to TA)	No breadth + persistence survivor net of fees	DEAD — didn't clear cost wall
OFI (order-flow)	Cont-style queue-change signal	Real micro-signal, but net-negative as taker; maker = adversely selected	DEAD — edge < spread
Lead-lag	Trade alt's under/over-shoot vs BTC-implied move	Raw BTC lead ≈ 0 (~ 15 s HFT game); residual catch-up failed OOS	DEAD — too fast / no edge
Funding carry	Delta-neutral long spot / short perp, collect funding	Headline $\sim 19\%$ /yr; our net after financing + fees + rotation = thin, tail-risky	SET ASIDE — compressing
Pairs / stat-arb	Cointegration mean-reversion, market-neutral	Beat passive on paper; our OOS weak — cross-sectional momentum weak	SET ASIDE — marginal after costs
Vol risk premium	Sell overpriced implied vol (short variance)	Real premium, but fat left tail; needs an options venue (Deribit)	SET ASIDE — un-built for our setup
Intraday-TSM	Morning→afternoon session momentum, daily hold	+16% on 1yr → -17% / Sharpe 0.22 / -52% DD on 5yr; selective 'careful-bets' version WORSE OOS	NOT CLOSED (live only) — both backtests failed; alive only as a live forward test
Stablecoin-flow	On-chain stablecoin supply timing (non-price)	Sharpe ~ 1.3 on 1yr → -8.4% / Sharpe -0.17 on 5yr vs buy-hold +40%	DEAD — failed 5yr; regime luck (IS -0.61 → OOS +0.32)

The directional-prediction family — everything that died the same death

Kronos — the original engine

A transformer price-predictor wrapped in a disciplined execution stack (regime sieves, confluence votes, ATR brackets, half-Kelly sizing). Beautifully built; it just couldn't predict.

- **Take-all baseline: 47–49% win**, Sharpe **–0.037 to –0.085** — a coin-flip that loses after bracket geometry and fees.
- Confident calls often flatly wrong — nights of **28–30 of 30 Monte-Carlo paths** pointing the wrong way.
- Lopsided noise by coin: **ADA shorts 7W/13L (35%)** vs ADA longs 7W/1L, BTC shorts 7W/2L.

Kelly / the meta-labeler — the filter that couldn't filter

- **v1 (101 trades)**: holdout **33.3%** vs a **61.9%** predict-majority baseline — worse than guessing the majority class.
- **Retrain**: ~45% vs ~51% baseline. Never beat it; stayed in **shadow**, never promoted to veto.

The entry-rule search — the seductive one

Four runs (A–D) with deflation guards (PSR, Deflated Sharpe). Every run a GREEN banner — a different winner each time: Run A funding+macd (OOS 0.066)→–0.03; Run B macd+rsi+stoch (OOS 0.267); Run C conviction+stoch+votes (0.090); Run D conviction+supertrend (0.092) but **86% IS win** → **53% OOS**. That collapse is the overfitting fingerprint; **conviction≥0.9 was already killed** by a per-coin test (BTC 15% win).

Consensus TA + RSI-2 reversion — both sides of the same coin

- 7-indicator trend board over **5 months: 48% win at 1:1** → **net loss** (price chops, not trends).
- The opposite (RSI-2 fade) also failed to produce a survivor net of fees. Neither trend nor counter-trend has a retail edge at this frequency.

OFI and lead-lag — 'real but unreachable'

- **OFI** carried information but is **net-negative as a taker**; the maker version is adversely selected.
- **Lead-lag**: raw BTC→alt lag is ~15s (HFT); the residual catch-up cousin had correlation ≈ 0 and failed IS/OOS.

The structural / market-neutral family — researched, mostly set aside

These collect a premium instead of predicting. More durable in principle, but each had a real blocker for *our* setup:

- **Funding carry**: headline ~19%/yr, <2% DD, but compressing as retail piles in; two legs + margin + borrow costs + liquidation tail-risk make the small-operator net far thinner.
- **Pairs / stat-arb**: beats passive in the literature; our OOS weak, cross-sectional momentum confirmed weak.
- **Vol risk premium**: robust premium but a **fat left tail** and needs an options venue (Deribit) we aren't on.

The reframe: **stop being the gambler, become the house** — DeFi LP vaults (HLP-style: ~22% CAGR, Sharpe 2.9–5.2), sell insurance (vol premium), or read on-chain information before it hits price.

The two that lasted longest — both failed the 5-year bar

Intraday-TSM was the lone directional edge to clear the cost wall in early testing: 1-year backtest \$5,000 → **\$5,799 (+16%)**. But the 5-year run told the truth: \$5,000 → **\$4,139 (–17%, CAGR –3.8%/yr)**, max DD **–52%**, Sharpe **0.22** — the +16% was carried almost entirely by one month (Jan 2026 +20%).

The 'careful bets' idea was tested and rejected. The hope: trade only the very biggest morning moves to concentrate a stronger edge. The selectivity ladder (top ■→¼→15%→10%→5%) showed OOS Sharpe getting **worse** as we got pickier (0.11 → –0.00 → 0.05 → –0.42 → –0.26). The mechanism is clean: **the biggest morning moves are usually overreactions (news shocks, liquidation spikes) that mean-revert** — so filtering for the biggest moves filters for the reversals. Selectivity is not the missing piece. Intraday-TSM is kept alive **only as a live testnet forward test**, not because the backtests are promising.

Stablecoin-flow also failed its 5-year test. On one year it looked strong (Sharpe ~1.3) — but that was beating a *negative* buy-and-hold in a flat year. Over five years: **-8.4% total, Sharpe -0.17, while buy-and-hold returned +40% (Sharpe 0.40)** — a market-timer that lost money *while the market rose*. The split confirms regime luck: **IS -0.61 → OOS +0.32**, with a single outlier month (2024-11, +12.1%) carrying most of the recent gain. 30 of 60 months positive is an exact coin flip. Real as *context* (supply contraction is mildly bearish), not a standalone edge.

Where that leaves us: every directional and timing road has now cleared exactly nothing on the 5-year, multi-regime, out-of-sample bar. The one family never disproved — because never built — is the structural '**be the house**' set: DeFi LP vaults and the volatility risk premium. They collect a premium for providing liquidity or insurance instead of predicting, so they don't hit the cost wall that killed everything here. That's the honest next road — same backtest-before-a-cent treatment as everything above.

The Simons lens — why fast retail prediction is structurally impossible

The famous '66%' is Medallion's **annual return, not its win rate** — their win rate was about **50.75%**. They won on a tiny per-trade edge (0.01–0.05%) × ~300,000 trades/day × ~12.5x leverage × **near-zero per-trade cost** (they're paid the spread). Our cost (**15–20 bps**) is **10–20x bigger** than the fast edges we chased — so fast directional scalping is closed to us by arithmetic. The only edges that can work are the inverse: slow, fatter-per-trade, structural.

The honest meta-lesson

The most reliable tell across all of this: **the in-sample → out-of-sample collapse.**

- Rule search: 86% → 53% win.
- Strategy search greens: 79–88% IS → 54–58% forward.
- Intraday-TSM: +16% cherry year → -17% over five.
- Stablecoin-flow: Sharpe 1.3 cherry year → -0.17 over five (IS -0.61 → OOS +0.32).

Every time we got excited, it was a small sample or an in-sample fit; every time we demanded honest OOS / forward / multi-regime evidence, the edge evaporated. **The discipline of refusing to trust the flattering number is the most valuable thing this project produced** — it's what keeps real money safe. The failures weren't wasted; they're the map of where the edge *isn't* — and they point clearly at the one road left: stop predicting, and collect a structural premium instead.

Internal sources: Kronos_Search_Runs_Tracker.md · CAPTAINS_LOG.md · Creative_Edges_Deep_Dive.md · Backup_Strategy_Options.md · consensus_backtest.py, reversion_backtest.py, ofi_maker_backtest.py, leadlag_scanner.py, funding_backtest.py, pairs_backtest.py, vol_premium_backtest.py, intraday_tsm_equity.py, intraday_tsm_careful.py, stable_flow_strategy.py (methods + verdicts).